

Practice for mini coding project 1

Part A: key skills in RStudio (8 marks)

Question 1 (3 marks total)

Write R code to achieve the following:

- a. 3 plus 1, all to the power of 3

Solution.

You need to use brackets around $3 + 1$ as otherwise with the order of operations R will do $3 + 1$ (raise 1 to the power of 3 first)

```
(3 + 1) ^ 3
```

```
[1] 64
```

- b. Finding the log (base 2) of 15

Solution.

We can use the `log()` function, and adjust the `digits` argument to 2 (if you forget the argument name, run `?log` in the console to bring up the help page). We could also use the `log2()` function instead:

```
# with log function  
log(15, base = 2)
```

```
[1] 3.906891
```

```
# with log2 function  
log2(15)
```

```
[1] 3.906891
```

c. Finding the type (class) of iris

Solution.

Recall that you find an object's type with the `class()` function:

```
class(iris)
```

```
[1] "data.frame"
```

Question 2 (3 marks total)

I've tried to write the following code, but it gives me an error:

```
2016_deaths <- c(100, 150, 40, 55)
```

```
Error in parse(text = input): <text>:1:5: unexpected input  
1: 2016_  
   ^
```

a. Explain in words why there is an error (1 mark)

Solution.

The error occurs because the variable name begins with a number. Numbers are allowed in variable names, but they can't be at the start

b. In the code chunk below, correct the mistake (2 marks)

Solution.

Something along the lines of:

```
deaths_2016 <- c(100, 150, 40, 55)
```

As long as the number isn't first, the variable name is ok.

Question 3 (2 marks)

I'm working with some mouse survival data in R. Two mice survived zero days, one survived 10 days and one survived 15 days. After collecting the data, I realised that the last mouse actually lived 16 days instead of 15, so I want to add one to that number. I've tried to do this as follows:

```
# the data I collected:
days_survived <- c("Zero", "Zero", 10, 15)

# actually, there has been a mistake! Need to add 1 to the last element
days_survived[4] <- days_survived[4] + 1
```

```
Error in `days_survived[4] + 1`:
! non-numeric argument to binary operator
```

I know that 15 + 1 should be 16, but this code gives an error.

- a. Explain why (1 mark)

Solution.

The original `days_survived` vector I defined contains a mix of character and numeric types. So, it gets coerced to character, because any number can be made into a character type but not all characters can be numbers (e.g. 15 can become "15" but "hello" isn't a number). If you check the type of days survived with `class()` or look in the environment panel you will see that this is a character type.

Because it's a character, `days_survived[4] + 1` gives an error as we're effectively doing "15" + 1 and we can't add a numeric type to a character type.

- b. In the code chunk below, correct the mistake (1 mark)

Solution.

You either need to wrap `days_survived[4]` in `as.numeric()`:

```
# the data I collected:
days_survived <- c("Zero", "Zero", 10, 15)

# actually, there has been a mistake! Need to add 1 to the last element
days_survived[4] <- as.numeric(days_survived[4]) + 1
```

Or alternatively change the "Zero" to be a number so the vector is numeric:

```
# the data I collected:
days_survived <- c(0, 0, 10, 15)

# actually, there has been a mistake! Need to add 1 to the last element
days_survived[4] <- days_survived[4] + 1
```

Part B: analysing biomedical datasets (12 marks)

Question 5 (1 mark)

How many rows and columns are in the `air_quality` dataframe? Hint: the help page for the `nrow()` function might be good to look at here.

Solution.

There are 153 rows and 6 columns.

To get this you could use the `nrow()` and `ncol()` functions like so:

```
nrow(air_quality)
```

```
[1] 153
```

```
ncol(air_quality)
```

```
[1] 6
```

Other acceptable answers include:

```
# the dim() function gives the number of rows and number of columns
dim(air_quality)
```

```
[1] 153  6
```

```
# the str() function describes the structure of a data frame (basically what
# appears in the environment panel)
str(air_quality)
```

```
'data.frame': 153 obs. of 6 variables:
 $ Ozone : int 41 36 12 18 NA 28 23 19 8 NA ...
 $ Solar.R: int 190 118 149 313 NA NA 299 99 19 194 ...
 $ Wind : num 7.4 8 12.6 11.5 14.3 14.9 8.6 13.8 20.1 8.6 ...
 $ Temp : int 67 72 74 62 56 66 65 59 61 69 ...
 $ Month : int 5 5 5 5 5 5 5 5 5 5 ...
 $ Day : int 1 2 3 4 5 6 7 8 9 10 ...
```

Or anything else that gives the correct answer through R code, as long as it is clear (through text or comments) which one is the rows and which one is the columns.

Question 6 (2 marks)

Let's start by exploring the data.

- a. How could you easily view `air_quality` as a table?

Solution.

You need to either use the `View()` function:

```
View(air_quality)
```

Or you could simply click on the little table button in the environment panel

- b. Write code to view the first 10 rows of `air_quality`

Solution.

You need to either use the `head()` function and set the `n` argument to 10 like so:

```
head(air_quality, n = 10)
```

	Ozone	Solar.R	Wind	Temp	Month	Day
1	41	190	7.4	67	5	1
2	36	118	8.0	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10

Question 7 (2 marks total)

The `air_quality` dataframe contains measurements of the air quality in New York City. One piece of information that has been recorded is the temperature each day.

- a. What is the range of temperatures?

Solution.

The range is between 56 and 97 degrees.

You first need to access the `Temp` column, using the `$` operator. Then, you could use the `range()` function:

```
range(air_quality$Temp)
```

```
[1] 56 97
```

Or you could use the `min()` and `max()` functions like so:

```
min(air_quality$Temp)
```

```
[1] 56
```

```
max(air_quality$Temp)
```

```
[1] 97
```

You need to be careful to use the correct case of the column name (i.e. `Temp` not `temp`)

- b. What is the average temperature?

Solution.

The average temperature is 77.88235 (also accept any rounding of this number e.g. 77 or 77.88)

You need to use the `mean()` function, with `$` to access the `Temp` column:

```
mean(air_quality$Temp)
```

```
[1] 77.88235
```

Question 8 (2 marks total)

- a. By default, this data is sorted by the Month column, and then Day column (with the first row being the 1st of May, and so on). What was the ozone level on the 22nd of May?

Solution.

The ozone level on the 22nd of May is 11.

Since we know the data is already sorted in the way we require, we could simply use indexing to find it:

```
air_quality$Ozone[22]
```

```
[1] 11
```

Or, we could use the `head()` function to show the first 22 rows, and read off this:

```
head(air_quality, n = 22)
```

	Ozone	Solar.R	Wind	Temp	Month	Day
1	41	190	7.4	67	5	1
2	36	118	8.0	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
9	8	19	20.1	61	5	9
10	NA	194	8.6	69	5	10
11	7	NA	6.9	74	5	11
12	16	256	9.7	69	5	12
13	11	290	9.2	66	5	13
14	14	274	10.9	68	5	14
15	18	65	13.2	58	5	15
16	14	334	11.5	64	5	16
17	34	307	12.0	66	5	17
18	6	78	18.4	57	5	18
19	30	322	11.5	68	5	19
20	11	44	9.7	62	5	20

21	1	8	9.7	59	5	21
22	11	320	16.6	73	5	22

Amongst many other solutions!

b. What are the ozone levels on the last three days of September?

Solution.

The ozone levels on the last three days of September are 14, 18 and 20.

There are two main ways we can achieve this. Firstly, we could use indexing on the Ozone column:

```
air_quality$Ozone[151:153]
```

```
[1] 14 18 20
```

Note that unlike in many other languages, a negative index in R removes that item from the vector, rather than allowing you to access elements from the end.

Or, we can use the `tail()` function:

```
tail(air_quality, n = 3)
```

	Ozone	Solar.R	Wind	Temp	Month	Day
151	14	191	14.3	75	9	28
152	18	131	8.0	76	9	29
153	20	223	11.5	68	9	30

Question 9 (1 mark)

Use the summary function to investigate the Ozone levels. Can you comment on the distribution of the data?

Solution.

Ozone levels range from 1-168, with 37 missing values. The average ozone level is 42.13 and the median is 31.5, so the distribution is positively (right) skewed.


```
summary(air_quality$Ozone)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NAs
1.00	18.00	31.50	42.13	63.25	168.00	37

Question 10 (4 marks total)

Let's take a look at the temperature column.

- a. It's currently a bit difficult to interpret the temperature values, as they're in Fahrenheit, not Celsius. We can convert them using the following formula: $C = (F - 32) * 5/9$. Add a new column called 'temp_c' with the temperature in Celsius. (2 marks)

Solution.

Remember we can create a new column using the assignment operator like so:

```
air_quality$temp_c <- (air_quality$Temp - 32) * 5/9
```

- b. What is the median temperature in Celsius over the study period? (1 mark)

Solution.

```
median(air_quality$temp_c)
```

[1] 26.11111

- c. On the 5th of May, the weather forecast predicted that the temperature would be 17 degrees Celsius. Was the actual temperature higher or lower than this? (1 mark)

Solution.

It was colder on the 5th of May than the forecast predicted.

We can answer this question using the comparisons covered in workshop 1, and our knowledge that the 5th of May is the 5th row of the data:

```
air_quality$temp_c[5] > 17
```

[1] FALSE

On the 5th of May, it was 13.3 degrees, so the above expression evaluates to FALSE.